The paleomagnetism of Keweenawan dikes from Baraga and Marquette Counties, northern Michigan

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A paleomagnetic study involving alternating field (AF) and thermal cleaning, baked contact tests, and the measurement of hysteresis and other magnetic properties has been carried out on 21 Keweenawan diabase dikes (age ~ 1.1 Ga) from Baraga and Marquette Counties, northern Michigan. The main results are as follows.

(1) The dikes exhibit a steep, negatively inclined remanence, which is shown to be a thermoremanent magnetization (TRM) acquired parallel to the ambient field at the time of initial magma cooling.

(2) A secondary component, more resistant to magnetic cleaning than the primary TRM, and with west-northwest declination and positive inclination is occasionally found in the dikes, especially those from Baraga County.

(3) Despite an early pole position obtained by Graham, which suggested that the dikes in Michigan were potentially important in defining the depth of the Logan Loop, most of them, from Marquette County, yield a pole that lies on the loop's western arm. The region of the projected apex of the loop thus still remains without reliable data.

(4) The pole position of the Marquette dikes (48.4°N, 213.5°E; $dp = 5.2°$; $dm = 6.2°$) obtained from 14 sites is virtually identical to that obtained from 17 reversely magnetized dikes from the Thunder Bay district in Ontario, but appears to be distinct from poles derived from other Keweenawan units. Both dike swarms are thus interpreted to represent parts of the same magmatic pulse, and one that occurred in the centre of Lake Superior during an early opening phase of the Keweenawan rift system.

On a fait une étude paléomagnétique comprenant le nettoyage magnétique dans un champ alternatif et par traitement thermique, des essais sur les contacts cuits, et la mesure de l'hystérèse et d'autres propriétés magnétiques sur 21 dykes de diabase du Keweenawien (datant d'environ 1.1 Ga) provenant des comtés de Baraga et de Marquette dans le nord du Michigan. En voici les principaux résultats:

(1) Les dykes montrent une rémanence abrupte inclinée négativement qui serait une aimantation thermique rémanente (ATR) qui s'est développée parallèlement au champ ambient au moment du refroidissement initial du magma.

(2) On retrouve occasionnellement une composante secondaire, plus résistante au nettoyage magnétique que l'ATR primaire, de déclinaison ONO et d'inclinaison positive surtout dans les dykes du comté de Baraga.

(3) En dépit d'une position de pôle précoce obtenue par Graham qui a suggéré que les dykes du Michigan étaient potentiellement importants pour définir la profondeur de la boucle de Logan, la plupart des dykes du comté de Marquette donnent un pôle qui repose sur la branche ouest de la boucle. Ainsi la région du sommet projeté de la boucle demeure toujours définie avec des données discutables.

(4) La position du pôle pour les dykes de Marquette (48.4°N, 215.5°E; $dp = 5.2°$, $dm = 6.2°$) obtenue à 14 sites est virtuellement identique à celle qui a été obtenue à partir de 17 dykes d'aimantation inversée dans le district de Thunder Bay en Ontario, mais elle semble être distincte des pôles dérivés pour d'autres unités du Keweenawien. Ainsi, on interprète les deux essaims de dykes comme représentant des portions de la même pulsation magmatique qui s'est produite au centre du lac Supérieur durant une phase précoce d'ouverture du système de rift keweenawien.

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Introduction

Paleomagnetic data from Late Precambrian Keweenawan (1.0–1.2 Ga) rocks of the Lake Superior region define a prominent loop in the North American apparent polar wander path, around which there is an anticlockwise movement of the pole (DuBois 1962; Robertson and Fahrig 1971; Spall 1971). This so-called "Logan Loop" (Robertson and Fahrig 1971) forms the polar wander tracks 3A and 3B of Irving and McGlynn (1976). It is defined by both normal and reversed poles,

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which are the consequence of at least two field reversals during Keweenawan time. Wherever found the reversals are always asymmetric with the downward (assumed normal) magnetization having an inclination 15–40° less than that of the reversed one. Due to this asymmetry the Logan Loop has large segments that are devoid of data, and thus its exact shape and nature are critically dependent upon the cause of the reversal asymmetry. Of particular concern therefore in the interpretation of Keweenawan paleomagnetism is whether the asymmetry is due to a secondary magnetization component (Palmer 1970), to apparent polar wander, (e.g., Beck 1970; Robertson and Fahrig 1971; Pesonen and Halls 1977) or to some fundamental difference in the geomagnetic field source between normal and reversed polarities (e.g., Robertson and Fahrig 1971; Pesonen 1978).

The western arm of the Logan Loop is well established by more than 15 reliable pole determinations (Pesonen 1978) whereas the eastern arm and northerly apex are by comparison poorly constrained (see Fig. 7). One of the few poles that appears to define the apex of the loop is that for the reversely magnetized Baraga County dikes from Michigan (pole BD in Fig. 7) obtained by Graham (1953). However this early study was performed on only two dikes without detailed magnetic cleaning. In view of the potentially key position of the Baraga pole a more comprehensive study of the Michigan dikes from both Baraga and adjacent Marquette Counties is warranted.

The purpose of the present study is twofold:

(1) To show if the paleomagnetic pole from the Baraga–Marquette dike swarm does indeed constrain the depth of the Logan Loop.

(2) To demonstrate that the reversed magnetization in these dikes is a thermoremanent magnetization (TRM) acquired at the time of initial cooling. The nature of the natural remanent magnetization (NRM) will be examined in particular detail as this paper and a companion one on the Thunder Bay dikes of Ontario (Pesonen 1979) form part of a larger paleointensity investigation of Keweenawan dikes and their baked country rocks (Pesonen 1978; L. J. Pesonen and H. C. Halls, in preparation). The demonstration of TRM is an essential requirement before reliable paleointensity estimates can be obtained.

Geological Setting

The area of study lies in the Upper Peninsula of Michigan in the Southern Province of the Canadian Shield (Fig. 1). The area is characterized by three basins of metasedimentary and metavolcanic rocks of Animikean age (2.1–1.8 Ga), which form part of the Marquette Range Supergroup (Cannon and Gair 1970), and which lie unconformably on a dominantly granitic basement of Archean age (Gair and Thaden 1968; Van Schmus and Woolsley 1975).

East–west-trending diabase dikes form an extensive swarm from Baraga County to the eastern margin of Marquette County (Balsley *et al.* 1949; Gair and Thaden 1968). The thickness of the dikes varies from a few centimetres up to 40 m. Their strikes vary from N50°E to N110°E with dips ranging from vertical to 75°N. The dikes are poorly exposed but can be easily traced for several tens of kilometres on aeromagnetic maps where they show up as spectacular linear negative anomalies (Balsley *et al.* 1949; Case and Gair 1965). The trend of these anomalies (and thus the general strike of the dikes) is parallel to the Lake Superior rift axis (Halls 1978).

Although there are no radiometric age data for the dikes, Aldrich *et al.* (1965) report a whole-rock K–Ar age of 960 Ma for an unmetamorphosed diabase dike in Iron County south of Baraga County. If this dike is coeval with the main swarm, then the Baraga–Marquette dikes are Keweenawan in age. The relative age of the dike swarm can be defined within broad limits based on geological field relations. They intrude both the Archean and Animikean rocks and show distinct chilled margins (Graham 1953; Gair and Thaden 1968; Cannon and Klasner 1972). Petrologically the dikes have not been affected by the last post-Animikean metamorphism at about 1.7 Ga ago (Van Schmus and Woolsley 1975). However, the Jacobsville sandstone, a flat-lying unit of probable Upper Keweenawan or Lower Cambrian age (Hamblin 1958; Halls and West 1971) overlies the Baraga–Marquette dikes at their western end. This relation is clearly shown by the aeromagnetic data as an increase in wavelength of dike anomalies on the western side of the contact between the Jacobsville and older rocks (Meshref and Hinze 1970). The dikes are therefore older than the Jacobsville and on the basis of Graham's (1953) paleomagnetic data are considered to be Lower Keweenawan in age (DuBois 1962; Books 1972; Table 6).

Cross-cutting relations suggest that there are two ages of Keweenawan dikes in the study area (Gair and Thaden 1968; Kantor 1969; L. Babcock, personal communication, 1976). The younger dikes (e.g., sites R2, R6, R13 in Fig. 1) are of a dark grey

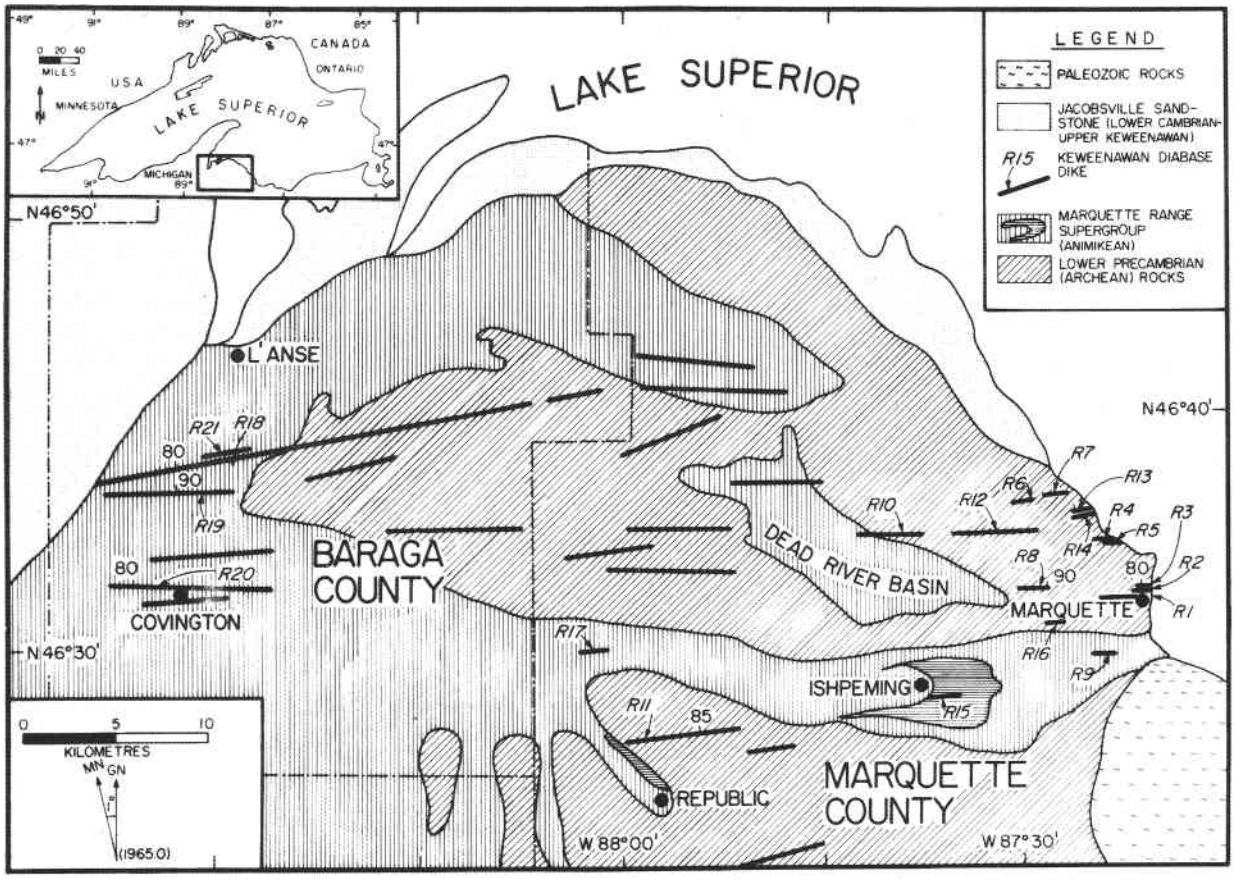


FIG. 1. Simplified geological map of Upper Peninsula of northern Michigan showing the east–west trending Baraga–Marquette dike swarm. The locations and extensions of the dikes have been determined using data from Gair and Thaden (1968), Cannon and Klasner (1972), Puffet (1974) and aeromagnetic maps by Meshref and Hinze (1970). Sampling sites (R1–R21) are indicated. The numbers near the dikes indicate their northerly dip, in degrees.

diabase. In places the diabase is porphyritic and on freshly broken surfaces the phenocrysts resemble fish scales (Gair and Thaden 1968). Plagioclase and pyroxene are the major minerals with subsidiary biotite, quartz, and olivine. Primary subhedral titanomagnetite is the main opaque mineral, although pyrite, ilmenite, and a coloured iron-oxide (presumably hematite) are occasionally present. Secondary alteration is minor but locally strong alteration has occurred (Gair and Thaden 1968; Puffet 1974). Common secondary minerals are chlorite, sericite, and epidote presumably formed as a result of deuteric alteration (Gair and Thaden 1968; L. Schutts, personal communication, 1977).

The older dikes (e.g., sites R4, R5, R14 in Fig. 1) are a more greenish diabase with a salmon-coloured groundmass. The dikes in Baraga County are slightly more altered than most of those in Marquette County and thus may belong to the older type of diabase.

Experimental Procedure

One hundred and thirty oriented hand samples were collected from 21 diabase dikes, 17 from Marquette County (sites R1–R17) and 4 from Baraga County (sites R18–R21). Orientation was carried out using a sun compass and checked with a magnetic compass. Five to ten samples were obtained from each dike (site). At three sites (R2, R5, R13) additional samples were collected from baked host rocks (Archean granite at sites R5 and R13 and Animikean metadiabase at site R2). Unbaked country rocks were also sampled at sites R2 and R13 in order to conduct a full baked contact test (Wilson 1962; Everitt and Clegg 1962).

Samples were drilled in the laboratory to obtain three cylindrical specimens with a height–diameter ratio of 2.25/2.50 cm (Noltimier 1971). Their NRM properties (Table 1) were measured with two commercial spinner magnetometers (PAR and DIGICO). The accuracies in paleomagnetic directions

TABLE 1. Magnetic properties of Marquette–Baraga dikes

Site	N	δ (g/cm^3)	χ (10^{-6}emu/cm^3	J_0 (10^{-6} emu/cm^3	Q	$H_{1/2}$ (Oe)	$T_{1/2}$ (°C)
			Marquette dikes				
R1	7	3.02	1863	19200	17.5	275	555
R2	6	3.01	1289	7350	8.4	390	530
R3	5	3.00	1321	16200	20.7	520	510
R4	5	2.89	2144	11700	11.3	270	—
R5	5	2.88	1127	4850	8.6	270	—
R6	5	2.96	893	7210	13.5	400	—
R7	4	3.01	483	2140	3.8	75	315
R8	5	2.99	1205	2580	4.4	230	325
R9	8	2.99	1112	1460	2.2	340	540
R10	6	3.09	1351	2830	3.5	370	505
R11	7	3.09	1983	5520	4.5	265	515
R12	5	2.99	659	2020	3.2	250	510
R13	4	2.94	1129	1770	2.4	330	490
R14	4	2.98	1089	2640	4.7	285	—
Mean Marquette	14	2.99	1260	6248	7.8	305	480
Standard deviation (%)		2	37	91	76	33	18
			Baraga dikes				
R18	5	3.06	2293	9750	7.1	325	510
R19	5	3.09	1823	11000	10.1	325	535
R20	5	3.09	1844	13800	12.7	280	535
R21	4	2.87	1241	5248	7.1	360	540
Mean Baraga	4	3.03	1800	9950	9.3	323	530
Standard deviation (%)		4	24	36	28	9	3

NOTES: Three sites (R15–R17) have been excluded from this and later analyses. The dike at R15 near Ishpeming (Fig. 1) has been affected by strong local magnetic fields due to an adjacent Animikean iron formation, and dikes R16 and R17 appear to have been broken up by late Keweenawan faulting (Klasner *et al.* 1975). Although individual samples from all three sites maintain stable NRM directions of AF cleaning, within-site dispersion is very high. N = number of samples; δ = density; χ = low field susceptibility; J_0 = NRM intensity; Q = Koenigsberger ratio ($J_0/H\chi$) where H is the present field strength in oersteds; $H_{1/2}$ and $T_{1/2}$ = median destructive field and temperature. Conversion of units from the cgs to the SI system: χ: 1 cgs unit = $1/4\pi$ SI units; J_0: 1 emu/cm^3 = 10^3A/m; H: 1 Oe = $10^3/4\pi$ A/m.

and intensities are estimated to be about $\pm 1°$ and $\pm 3\%$ respectively. Two hundred and forty specimens were subjected to alternating field (AF) demagnetization in constant steps of 50–100 Oe (4–8 × 10^3 A/m) to a maximum field of 1000 Oe (8 × 10^4 A/m) using a Schonstedt single-axis demagnetizer. Twenty-three specimens were thermally cleaned in steps up to 620°C. Temperature intervals ranged from 100°C at low temperatures to 5°C at higher temperatures where the intensity falls rapidly. Hysteresis properties were measured with a PAR vibrating sample magnetometer. Anhysteretic remanent magnetization (ARM) for use in Lowrie–Fuller (1971) tests was induced using the AF demagnetizer coil around which a set of Helmholz coils was built to provide a direct current (DC) field of 0.5–1.2 Oe (39.8–95.5 A/m). For a more detailed description of measurement procedure and laboratory instruments, see Pesonen (1978).

AF Demagnetization Studies

AF demagnetization data from 240 specimens were carefully examined for systematic directional swings characteristic of superimposed remanence components. An optimum cleaning field (H_{opt}) and corresponding paleomagnetic stability index (PSI) were determined for each specimen following the method of Symons and Stupavsky (1974).

The majority (75%) of Baraga–Marquette dike specimens are directionally stable (i.e., PSI values less than 20) and yield a single characteristic remanence direction of reversed polarity with a steep upward inclination. Two examples of this behaviour are shown in Fig. 2. Both specimens yield well-defined optimum cleaning fields of 600 Oe (4.8 × 10^4 A/m) and 300 Oe (2.4 × 10^4 A/m) with corresponding PSI values of 4.3 (very stable) and 8.4 (stable), respectively. There is no evidence in either specimen of a directional swing in peak AF's up to 1000 Oe (8 × 10^4 A/m). In the Baraga dike

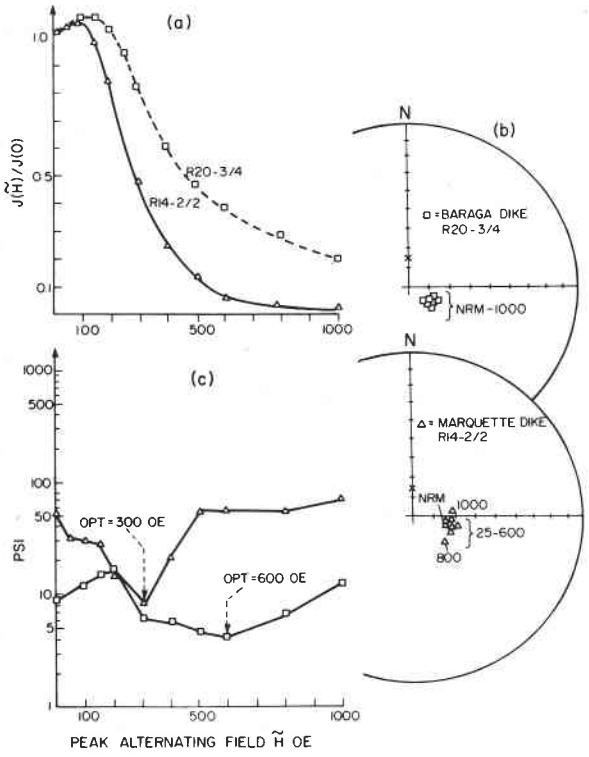


FIG. 2. Two examples of stable NRM upon AF demagnetization of Baraga–Marquette samples. (*a*) Normalized intensities. (*b*) Directions of NRM on a Lambert equal-area stereonet with open symbol indicating upward direction. (*c*) The paleomagnetic stability indices (PSI) of Symons and Stupavsky (1974).

specimen about 20% of the NRM remains after the final cleaning step whereas the Marquette specimen is virtually demagnetized. In both specimens a viscous component (VRM) due to the present Earth's magnetic field (PEF) is probably present as reflected in the low field data by an increase in intensity and a decrease of the PSI value with increasing peak alternating field (Fig. 2). Directional changes due to this low coercivity component are absent, however, presumably due to its relatively small intensity and also because the PEF direction is almost antiparallel to that of the characteristic reversed remanence. Some specimens, notably from sites R7 and R11, show much stronger viscous overprinting by the PEF; they are magnetically much softer and exhibit larger directional swings (10–150°). About 10% of the specimens show generally very small (3–10°) swings of random azimuth whose sense of movement is away from the characteristic direction. Although specimens from a single sample show similar demagnetization paths, the swings obtained from different samples (either from the same or a different site) are random. The cause of this behaviour is unknown but if it is due to a single harder component superimposed on the characteristic one, the directions of the two components must be almost antiparallel. If a single component overprint is involved it may arise from a field reversal or through recent weathering in the PEF (e.g., Irving 1964, p. 99). Laboratory experiments on the Michigan dikes using partial thermoremanent magnetizations suggest that a component due to self-reversal is unlikely (Pesonen 1978).

Only four specimens, from a single sample at site R4, showed any definite evidence for a secondary component other than that due to the PEF. All the specimens showed smooth great circle demagnetization paths away from the characteristic direction and reached a mean stable end point ($D = 298°$; $I = +2°$), which is very similar to other recently discovered Late Keweenawan secondary overprint directions (e.g., Henry *et al.* 1977; Palmer *et al.* 1976). However a positive baked contact test at the adjacent site R5 shows that the secondary overprint at site R4 has been acquired locally and does not represent a pervasive regional remagnetization event (see Fig. 3).

Thermal Demagnetization

Of the 23 specimens subjected to thermal cleaning, all of the Marquette dikes and most of the Baraga ones are directionally stable throughout the full range of blocking temperatures. Two examples are shown in Fig. 3. Because the thermal increments (ΔT) for any one specimen are highly variable, the variation in PSI values becomes sensitive not only to directional changes but also to changes in ΔT (Pesonen 1978). Hence a different method from that used for the AF data was adopted to define the optimum cleaning temperature. Running averages of remanence directions of three successive temperature steps were obtained and the circular standard deviation ($\theta 63$) was calculated for each average. Values of $\theta 63$ were then plotted against the midpoint of the temperature interval defined by each triplet of consecutive readings. The temperature at which this curve reaches a minimum then defines the optimum cleaning value (see Table 4) (Pesonen 1978).

Systematic westward swings, exhibited by six Baraga dike specimens, begin at temperatures usually above 450°C and can often only be detected using temperature intervals as small as 5°C. Three examples of swinging behaviour are shown in Fig. 4. Although no stable endpoints corresponding to the secondary component could be defined the demagnetization paths in Fig. 4 suggest that this component has a declination of about 290° and an inclination of at least 30° (c.f. Vincenz and Yaskawa

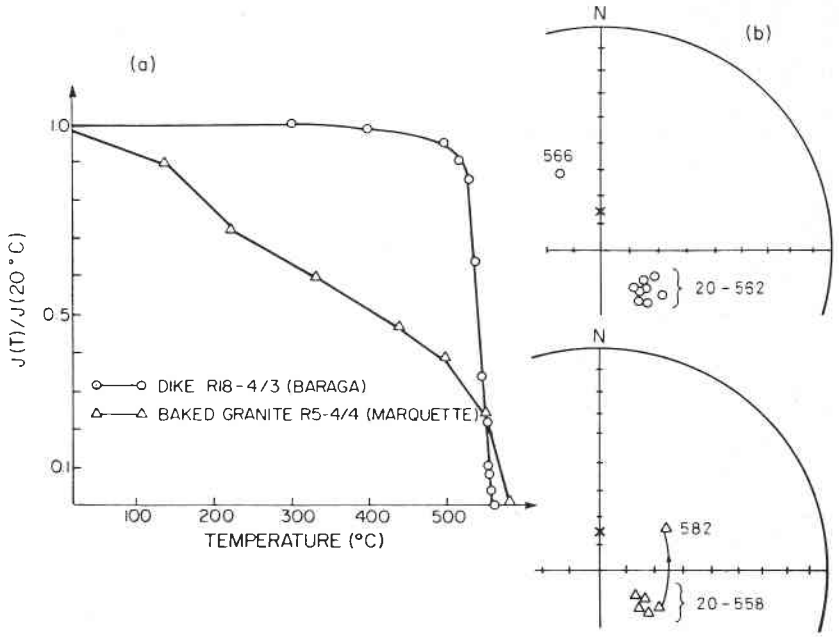


FIG. 3. Two examples of thermal demagnetization of Baraga–Marquette samples. Both specimens exhibit a stable NRM with steep negative inclinations. Note that the baked contact specimen (R5–4/4) yields the same characteristic direction of NRM as the dike (specimen R18–4/3) although their blocking temperature spectra are dissimilar.

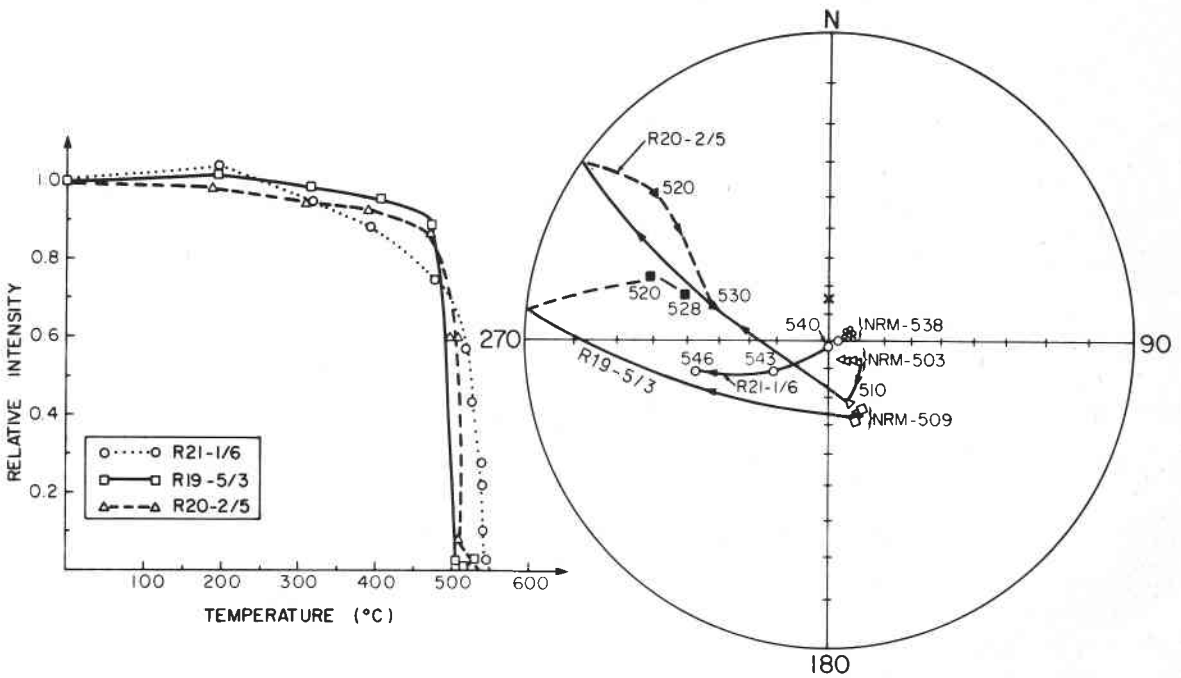


FIG. 4. Examples of directional swings of NRM's of three Baraga dike specimens upon stepwise thermal demagnetization. Open symbols indicate upward directions, closed symbols downward directions. Note that all specimens reveal a similar trend towards a secondary remanence component with a westerly declination and downward inclination.

TABLE 2. Results of baked contact tests on Marquette dikes

Rock type	N/n	D (deg)	I (deg)	k	α_{95} (deg)	χ (10^{-6} emu/cm^3)	Q
Mean of dikes	3/15	114	−66	47	18	1214	6.5
Mean of baked rocks	3/11	113	−66	119	11	27	6.2
Unbaked granite	1/1	209	28	—	—	3	3.1
Unbaked metadiabase	1/1	257	64	—	—	152	0.2

NOTES: N/n = (number of dikes)/(samples); D and I = declination and inclination of NRM; k and α_{95} = precision parameter and radius of 95% confidence circle about the mean direction (Fisher 1953); χ and Q: see Table 1.

1968 for similar results on Baraga dikes). The observation that all the trends have a similar azimuth suggests that the component is not antiparallel to the characteristic one.

Baked Contact Tests

In previous sections we showed that with the exception of one sample, AF and thermal cleaning were unable to disclose any evidence of a secondary component in Marquette dikes. However, it is conceivable that the secondary component has exactly the same coercivity and blocking temperature spectra as the characteristic component. This possibility can be investigated indirectly by the use of baked country rocks. Since these rocks (Archean granite and Animikean metadiabase) exhibit completely different lithology and oxide petrology compared to the dikes as reflected in their magnetic properties (Tables 1, 2), it might be expected that the dike and baked rocks have acquired dissimilar proportions of the secondary component resulting in a negative baked contact test.

The results, however, clearly demonstrate a fully positive baked contact test as the NRM directions of the igneous and baked rocks agree closely whereas those from the unbaked rocks are significantly different (Table 2). A feature that appears to be general for all baked contact tests carried out on Keweenawan rocks (Pesonen 1979) is that the directional dispersion in the baked rocks is less than that in the dikes (see k values in Table 2). Pesonen (1978) has shown that this difference is due to the higher paleomagnetic stability of baked specimens, which emphasizes their particular suitability for paleomagnetic studies. An additional feature is that Q, the Koenigsberger ratio, is much higher in the dikes and baked contact rocks than it is in the unbaked rocks (Table 2). This suggests that the NRM of the dikes and baked rocks is a primary TRM acquired during initial cooling subsequent to intrusion, and that the remanence in the unbaked rocks is not a TRM (Everitt and Clegg 1962). The positive baked contact test and also the laboratory-induced partial thermoremanent magnetization (PTRM) studies (Pesonen 1978) indicate that the reversed NRM in the Baraga–Marquette dikes defines the direction of a Late Precambrian paleofield (DuBois 1962; Vincenz and Yaskawa 1968) and is not due to a self-reversal (Graham 1953).

Rock Magnetic Studies

From the viewpoint of Thellier paleointensity studies (Pesonen 1978; L. J. Pesonen and H. C. Halls, in preparation) it is important not only to demonstrate the existence of a TRM but also to establish the domain state of remanence carriers (e.g., Levi 1977). This latter aspect was examined for the Baraga–Marquette dikes through measurement of hysteresis properties and the use of the Lowrie–Fuller test (Lowrie and Fuller 1971).

Hysteresis Properties

According to the results of Day *et al.* (1977) the mean J_{sr}/J_s and H_{cr}/H_c values of Marquette dikes (0.22; 1.85) and baked rocks (0.14; 2.37) given in Table 3 suggest that the remanence is carried by pseudo single-domain (PSD) grains. The Baraga dikes have slightly higher average J_{sr}/J_s values (0.31) than the Marquette dikes (0.22) indicating greater magnetic stability, a result that is consistent with their high median destructive field and temperature ($H_{\frac{1}{2}}$, $T_{\frac{1}{2}}$) and their higher Q values (Table 1). Two Curie-point determinations were made using the J_s–T method. Curie temperatures of 585 and 570°C were obtained for a Baraga and a Marquette dike respectively, indicating in both instances the presence of (almost) pure magnetite and a lack of hematite. Therefore the considerable variations in J_s and J_{sr} and also in low field susceptibility values (χ) found in Baraga–Marquette dike specimens (Tables 1 and 3) are thought to reflect the variable magnetite content of these rocks.

Lowrie–Fuller Tests

A comparison of normalised AF demagnetization curves of low field TRM with saturation isothermal remanent magnetization (IRM) can be used as an indicator of the domain state of rema-

TABLE 3. Hysteresis properties of Baraga–Marquette dikes

Specimen	J_s (emu/cm³)	J_{sr} (emu/cm³)	J_{sr}/J_s	H_c (Oe)	H_{cr} (Oe)	H_{cr}/H_c	T_c (°C)	Domain state
			Marquette dikes					
R1-3/2	7.52	2.34	0.31	302	485	1.61		PSD
R2-5/4	4.32	0.84	0.19	190	396	2.08	570	PSD
R3-3/3	3.47	1.18	0.34	315	538	1.71		PSD
R4-1/1	5.49	1.14	0.21	150	268	1.79		PSD
R4-4/3*	0.28	0.04	0.14	135	320	2.37		PSD
R7-3/2	5.87	1.09	0.19	175	262	1.50		PSD
R9-4/3	2.43	0.47	0.19	150	267	1.78		PSD
R10-6/2	3.98	1.19	0.30	261	393	1.51		PSD
R11-5/3	6.62	1.60	0.24	57	119	2.09		PSD
R11-1/1	0.21	0.03	0.15	95	181	1.91		PSD
R13-5/3	2.76	0.51	0.19	150	300	2.00		PSD
			Baraga dikes					
R18-3/1	7.82	2.36	0.30	267	406	1.52		PSD
R18-4/1	5.89	1.97	0.33	340	667	1.96		PSD
R19-3/2	5.76	1.64	0.29	225	328	1.46	585	PSD
Mean Marquette	3.91	0.95	0.22	180	320	1.85		PSD
Mean Baraga	6.50	1.99	0.31	277	467	1.65		PSD

NOTES: J_s = saturation magnetization; J_{sr} = saturation remanent magnetization; H_c = coercive force; H_{cr} = coercivity of remanence; T_c = Curie temperature; PSD = pseudo single-domain remanence carriers.
*Baked Archean granite.

nence carriers (Lowrie and Fuller 1971). If the NRM (assumed to be a TRM) is magnetically harder than the IRM, the domain state is single domain (SD), whereas if the NRM is softer than the IRM it is multidomain (MD). Furthermore a comparison of normalised AF demagnetization curves of NRM with those of laboratory-induced TRM and ARM gives insight into whether the NRM is of thermal origin (NRM, TRM, and ARM curves are similar) or of viscous or chemical origin in which case the NRM curve differs from those of TRM and ARM (Dunlop *et al.* 1973; Buchan 1977). An example of a Lowrie–Fuller test on a Marquette dike specimen from site R2 is shown in Fig. 5*a*. The ARM induced before heating (ARM1) and the TRM decay curves are very similar to that of the NRM suggesting that the NRM is a TRM (e.g., Levi and Merrill 1976). The TRM is, however, slightly harder than the NRM but this is likely due to physico–chemical changes during heating because the ARM induced after heating (ARM2) is much harder than the ARM1. The IRM is clearly the softest remanence indicating SD-like remanence carriers. This conclusion is supported by the observation that the ARM acquired with a DC field of 1.2 Oe (95.5 A/m) is softer than the ARM produced with a DC field of 0.5 Oe (39.8 A/m) (Dunlop *et al.* 1973).

For a country rock specimen baked by the dike R2 (Fig. 5*b*), the NRM and ARM1 have similar AF characteristics, indicating that the NRM is a TRM acquired during baking. Similar NRM–ARM relations are also found for a dike at site R21 (Fig. 5*c*). In contrast, a nonbaked rock from site R2 (Fig. 5*d*) has a NRM that is much softer (and weaker) than the ARM indicating that the NRM is not a TRM. This result further demonstrates that only the dikes and their adjacent baked rocks have acquired a TRM during cooling. The Lowrie–Fuller and hysteresis data suggest that this TRM is carried by PSD or SD magnetite grains and therefore that these rocks are suitable material for paleointensity studies (Levi 1977; Pesonen 1978).

Paleomagnetic Results

An optimum direction of NRM was selected for each specimen according to the methods outlined previously. Standard Fisher statistics (Fisher 1953) were then used hierarchically (specimens, samples, sites) to obtain group mean directions of NRM. The Marquette dikes are treated separately from the Baraga dikes because the latter yield systematically steeper inclinations of characteristic remanence and also appear to have suffered a partial magnetic overprint. The site mean paleomagnetic data for the Marquette dikes obtained either by thermal or AF demagnetization show close agreement (Table 4). The lack of directional streaking in the stereoplot of Fig. 6 is further evidence that only one characteristic component is present.

The mean direction of the Baraga dikes differs

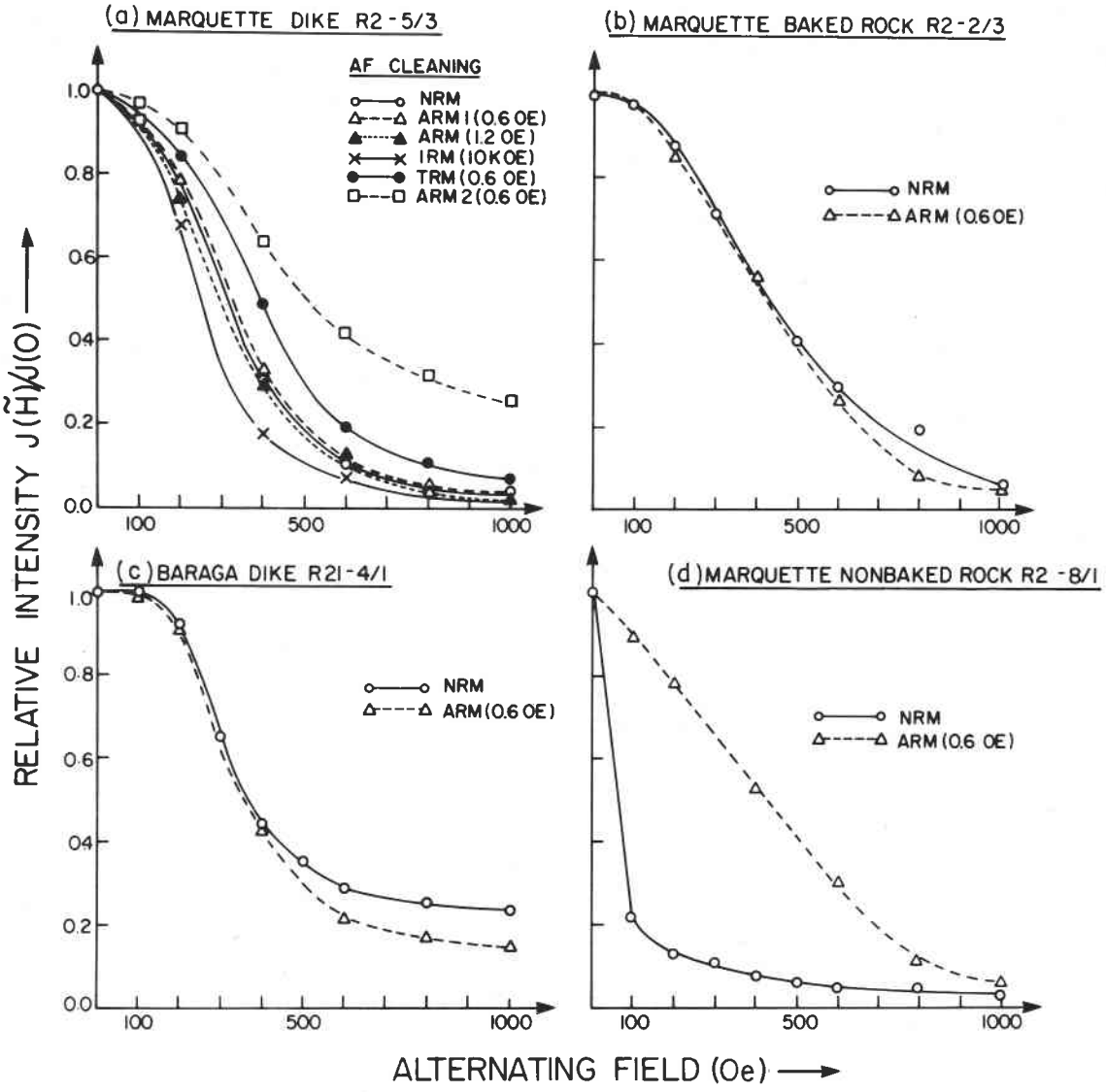


FIG. 5. Examples of Lowrie–Fuller test results. (*a*) Marquette dike specimen (R2–5/3). (*b*) Marquette baked county rock (R2–2/3, Animikean metadiabase). (*c*) Baraga dike specimen (R21–4/1). (*d*) Marquette unbaked county rock (R2–8/1). A full Lowrie–Fuller test is shown for specimen (*a*); for remaining specimens only NRM and ARM data are presented.

significantly ($P = 0.95$) from that of the Marquette dikes (Table 4). Three possible explanations may account for this difference:

(1) A high proportion of Baraga specimens contain a secondary component with westerly declination. Table 1 shows that the swinging specimens (e.g. from dike R21) also tend to have higher $H_{1/2}$ and $T_{1/2}$ values than stable specimens such as are found in dike R18. These results could suggest that the characteristic component obtained from some of the Baraga dikes may be biassed to steeper negative inclinations by failure to remove the secondary component in the cleaning process. It is possible in view of their greater degree of alteration that the Baraga dikes represent a deeper erosional level than the Marquette dikes and that they have been selectively overprinted during uplift, or that a slightly different lithology has rendered them more susceptible to partial remagnetization.

(2) Some Baraga samples (e.g., from site R20 in Fig. 2) show no evidence of a secondary component, and thus the steeper inclination of their pri-

TABLE 4. Paleomagnetism of Baraga–Marquette dikes

	AF cleaning							Thermal cleaning					
Site	AF (Oe)	PSI	N	D (deg)	I (deg)	k	α_{95} (deg)	T (°C)	N	D (deg)	I (deg)	k	α_{95} (deg)
					Marquette dikes (46.7°N, 87.5°W)								
R1	200–500	16.0	7	111.2	−71.1	96	6.2	506	1	119.4	−68.4		
R2	200–800	8.0	6	114.7	−68.4	210	4.6	350	4	114.7	−68.0	32	5.1
R3	100–600	6.5	5	99.3	−69.8	197	5.5	370	1	85.4	−58.7		
R4	100–60	28.3	5	136.5	−68.2	17	18.8	—	—	—	—		
R5	100–1000	14.8	5	120.5	−58.2	121	7.1	438	1	124.1	−68.0		
R6	100–1000	8.5	5	106.5	−67.3	41	12.1	—	—	—	—		
R7	300–1000	27.9	4	132.9	−70.0	822	3.2	286	1	129.3	−66.2		
R8	200–400	24.2	5	98.3	−76.4	541	3.3	325	1	64.5	−82.0		
R9	150–500	25.5	8	104.0	−64.4	98	5.6	325	1	80.4	−54.5		
R10	150–600	25.2	6	129.0	−53.1	46	10.0	400	1	121.3	−61.8		
R11	50–1000	21.8	7	114.5	−71.1	63	7.7	250	1	134.1	−81.3		
R12	200–800	10.6	5	113.6	−64.7	97	7.8	325	1	123.6	−61.9		
R13	0–400	24.0	4	103.2	−70.3	144	7.6	260	1	85.9	−67.9		
R14	200–1000	15.0	4	110.3	−73.8	446	4.3	—	—	—	—		
Mean Marquette dikes		18.3	14	115.0	−68.0	115	3.7		11	107.4	−68.4	48	6.6
					Pole position: 48.4°N, 213.5°E (dp = 5.2°; dm = 6.2°)								
					Baraga dikes (46.6°N, 88.5°W)								
R18	200–1000	14.3	5	104.8	−73.9	122	7.0	380	3	130.5	−76.7	198	8.8
R19	200–800	11.4	5	95.8	−78.4	34	13.4	348	3	179.4	−79.2	11	40.1
R20	200–1000	14.7	6	121.2	−76.8	730	2.8	396	2	130.8	−80.1	—	—
R21	0–400	12.7	4	111.9	−84.1	509	4.0	507	1	68.2	−83.7	—	—
Mean Baraga dikes		13.3	4	108.0	−78.4	281	5.4		4	134.9	−81.6	129	8.1
					Pole position: 48.9°N, 238.2°E (dp = 9.6°; dm = 10.2°)								

NOTES: AF = range of optimum specimen cleaning fields; PSI = paleomagnetic stability index (Symons and Stupavsky 1974); N = number of samples; D, I, k, α_{95}: see Table 2; T = optimum specimen cleaning temperature; dp and dm = semi-axes of the 95% confidence oval about the pole.

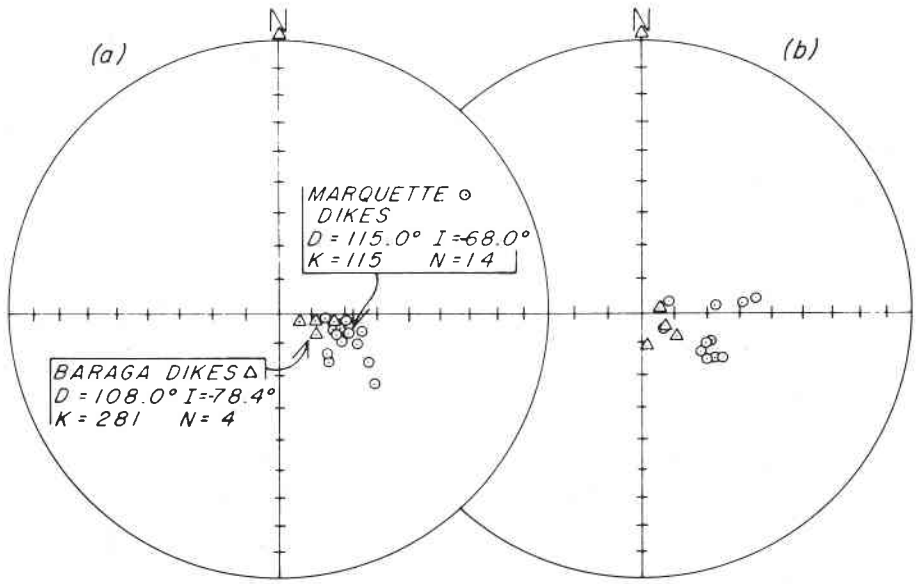


FIG. 6. Site-mean paleomagnetic directions of Baraga–Marquette dikes after (*a*) AF cleaning and (*b*) thermal cleaning. Lambert equal-area stereonets. Symbols as in Fig. 2.

TABLE 5. Comparison of paleomagnetic directions and magnetic properties of two Keweenawan dike swarms on opposite sides of Lake Superior

Dike swarm	Magnetic properties							Hysteresis data		Paleomagnetic data		
	N	δ (g/cm^3)	χ (10^{-6} emu/cm^3)	NRM	Q	$H_{1/2}$ (Oe)	$T_{1/2}$ (°C)	J_{sr}/J_s	H_{cr}/H_c	D (deg)	I (deg)	k
Marquette dikes	14	2.99	1260	6248	7.8	305	480	0.22	1.85	115.0	−68.0	115
Thunder Bay dikes*	17	2.95	1195	4119	6.4	249	483	0.22	2.19	112.3	−68.4	53

NOTES: N = number of sites; remaining symbols as defined in Tables 1–3.
*Data from Pesonen (1979).

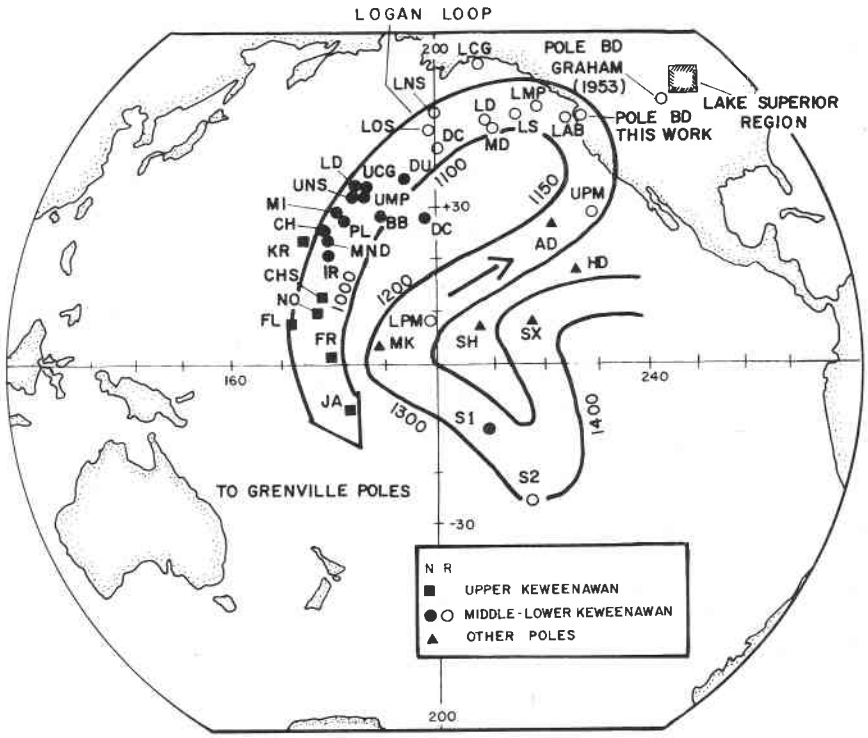


FIG. 7. Summary of the Logan Loop as recorded in Keweenawan rocks from the Lake Superior region. The Keweenawan poles are listed in Table 6. Other poles:+ AD = Aillik dikes;+ MK = MacKenzie Interval mafic igneous rocks;+ SH = Sherman Granite;+ SX = Sioux quartzite; HD = Harp dikes (Irving *et al.* (1977). Results with a (+) are taken from Irving and Hastie (1975).

mary component may indicate that they represent a slightly different (and perhaps older) magmatic episode than the Marquette dikes.

(3) The Baraga dikes are only about 5 km to the east from Keweenawan volcanics that strike N20E and dip 10–15° WNW. A structural correction to restore the volcanics to horizontal would move the Baraga pole in close coincidence with the Marquette pole.

The mean directions of the Marquette and Baraga dikes differ from those obtained by Graham (1953). In the case of the Marquette dikes this difference is significant ($P = 0.95$) but for the Baraga dikes it is less meaningful because only four dikes were sampled and one of them (R21) gives a direction very similar to that reported by Graham.

Discussion and Conclusions

The paleomagnetic poles from the Baraga and Marquette dikes (BD, MD) have been plotted on the Logan Loop in Fig. 7, which also shows other Keweenawan paleopoles as summarized in Table 6. The Baraga pole is close to that originally obtained by Graham (1953) at the apex of the loop but may be suspect due to the paucity of data and possible contamination by a secondary component. The

TABLE 6. Summary of Keweenawan (1300–1000 Ma) paleomagnetic poles from the Lake Superior region

	Rock unit	Age* (Ma)	P	S/B	Paleomagnetic pole Lat. (°N)	Long. (°E)	dp (deg)	A_{95} (deg)	dm (deg)	Ref.†
				Upper Keweenawan						
JA	Jacobsville sandstone	1000	N	1/24	−9.0	183.0	3.0		6.0	1
FR	Freda sandstone	1046	N	1/85	1.2	179.5	1.4		2.8	2
FL	Fond du Lac Formation		N		8.0	173.0				3
NO	Nonesuch shale	1060	N	1/27	10.3	175.6	3.1		6.2	2
CHS	Copper Harbor conglomerate secondary component		N	1/3	14.0	179.0				4
KR	Kearsage rhyolite	1007	N	1/6	25.4	170.7	5.0		9.4	5
				Middle Keweenawan						
CH	Copper Harbor lavas	1080	N	3/17	24.2	177.9		(15.2)		6, 7, 8
MND	Basaltic dikes, Minnesota		N	1/12	25.4	176.0	6.4		11.4	9
PL	Portage Lake lavas	1100	N	4/76	26.6	181.7	1.7		2.9	5, 6
IR	Isle Royale lavas		N	1/4	20.6	176.3	5.1		9.5	5
UNS	Upper North Shore lavas		N	3/98	32.4	183.8		(4.3)		5, 10, 11
MI	Michipicoten Island lavas	953	N	1/28	28.7	179.0	5.2		8.7	10
UCG	Upper Cape Gargantua lavas	1090	N	2/14	33.0	182.8		(12.2)		10, 12
UMP	Upper Mamainse Point lavas	1070	N	2/66	33.1	181.5		(16.7)		10, 12
LD	Logan dikes	1050	N	3/47	34.6	181.1		(9.1)		7, 8, 13, 14
BB	Beaver Bay Complex	1060	N	2/29	27.5	189.5	3.5		5.5	15
DU	Duluth Complex (Duluth area)	1115	N	4/70	36.2	192.0	5.6		8.1	16
DC	Duluth Complex (Cook Co.)		N	2/18	29.0	196.9		(7.5)		16
				Lower Keweenawan						
LNS	Lower North Shore lavas		R	3/38	49.8	199.2		(11.0)		10, 11, 17
LOS	Lower Osler lavas		R	2/37	46.1	198.7		(18.4)		10, 18
LCG	Lower Cape Gargantua lavas		R	2/11	59.9	212.9		(59.5)		10, 12
LMP	Lower Mamainse Point lavas		R	2/30	50.3	226.0		(46.1)		10, 12
LAB	Lower Alona Bay lavas		R	1/11	47.4	232.7	17.1		18.5	10
LS	Logan sills	1160	R	5/62	49.0	220.3		(4.1)		7, 8, 10, 14
LD	Logan dikes		R	1/17	48.0	212.2	7.0		8.3	8, 14
MD	Marquette County dikes		R	1/14	48.4	213.5	5.2		6.2	8, this work
BD	Baraga County dikes		R	1/4	48.9	238.2	9.6		10.2	8, this work
BD	Baraga County dikes		R	1/2	45.2	261.0	2.0		2.0	19
DC	Duluth Complex (Cook Co.)		R	1/13	42.5	203.5	7.0		9.5	16
UPM	Upper Powdermill lavas	1200	R	1/14	28.9	232.4	7.3		8.3	17
LPM	Lower Powdermill lavas	1220	N	1/7	9.8	199.6	7.8		13.5	5
S1	Sibley Series	1370	N	1/2	−13.0	210.8	8.0		16.0	20
S2	Sibley Series	1370	R	1/5	−24.0	218.0	10.0		19.0	20

NOTES: Mean paleomagnetic poles are given for units where several studies of comparable detail have been made. A more complete listing of Keweenawan poles and relevant statistical data is given in Pesonen (1978). P = magnetic polarity; N and R = normal and reversed; S/B = (number of studies)/(total number of sites); Lat., Long., dp, dm = latitude, longitude, and 95% confidence limits on the pole position; A_{95}: 95% confidence circle for a mean pole position.

*Further details and discussion of ages assigned to Keweenawan units can be found in Irving and McGlynn (1976), Henry *et al.* (1977), and Pesonen (1978).

†REFERENCES: 1 = Roy and Robertson (1978); 2 = Henry *et al.* (1977); 3 = Watts and Van der Voo (1977); 4 = Palmer *et al.* (1976); 5 = Books (1972); 6 = Vincenz and Yaskawa (1968); 7 = DuBois (1962); 8 = Pesonen (1978); 9 = Beck (1969); 10 = Palmer (1970); 11 = Hubbard (1971); 12 = Robertson (1973*b*); 13 = Robertson and Fahrig (1971); 14 = Pesonen (1979); 15 = Beck and Lindsley (1969); 16 = Beck (1970); 17 = Books (1968); 18 = Halls (1974); 19 = Graham (1953); 20 = Robertson (1973*a*).

Marquette pole plots on the western arm of the Logan Loop, thus leaving the apex still undefined by reliable data.

The Marquette pole is virtually identical to that found for reversely magnetized dikes (LD in Fig. 7) from the Thunder Bay area of Ontario (Pesonen 1979). This coincidence suggests that both dike sets represent parts of the same magmatic pulse that occurred in the centre of Lake Superior during early opening of the Keweenawan rift system. This conclusion is further supported by the fact that the physical properties of the two swarms in Table 5 are very similar. Both the Marquette and Thunder Bay poles appear to be distinct from poles obtained from other Keweenawan units around Lake Superior. In particular the Marquette dike pole

differs significantly from that of the reversed lavas of the Powdermill Group on the eastern arm of the Logan Loop (UPM in Fig. 7), suggesting that the Baraga–Marquette dikes are not feeders for these lavas. This conclusion is supported by geological and aeromagnetic data (Hubbard 1975; Meshref and Hinze 1970), which suggest that at least some of the Baraga dikes cut the Powdermill lavas.

The reversed pole of the Marquette dikes plots about 30°NE from the Middle Keweenawan normal poles (Fig. 7). The lack of a regional secondary component, particularly in the Marquette dikes, demonstrates that the asymmetry of the Keweenawan reverse-to-normal polarity change is not caused by a secondary component as originally suggested by Palmer (1970).

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